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Inventor(s) Gontcharov; Vladimir et al.

Loudspeaker arrangement

Abstract

A loudspeaker arrangement (**800**) having first and second loudspeakers mechanically coupled to each other. Either a first side of a first sound radiating surface faces a first side of a second sound radiating surface, and the first side of the first sound radiating surface and the first side of the second sound radiating surface both face a first cavity having a first opening facing towards an internal air volume, or a second side of the first sound radiating surface faces a second side of the second sound radiating surface, the first side of the first sound radiating surface faces a second cavity having a second opening facing towards the internal air volume, and the first side of the second sound radiating surface faces a third cavity separate and distinct from the second cavity, the third cavity has a third opening facing towards the internal air volume.

Inventors: **Gontcharov; Vladimir (Budapest, HU), Lahti; Hans Erik Otto (Torslanda, SE), Sterling; Brian (Farmington Hills, MI)**

Applicant: **Harman Becker Automotive Systems GmbH** (Karlsbad, DE)

Family ID: **75581524**

Assignee: **Harman Becker Automotive Systems GmbH (Karlsbad, DE)**

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Primary Examiner: Nguyen; Sean H

Background/Summary

CROSS REFERENCE (1) This application is a continuation of International Application No. PCT/EP21/059953, filed with the European Receiving Office on Apr. 16, 2021, entitled LOUDSPEAKER ARRANGEMENT, the disclosure of which is incorporated in its entirety by reference.

TECHNICAL FIELD
(1) The disclosure relates to a loudspeaker arrangement, in particular to a loudspeaker arrangement within a housing.
BACKGROUND
(2) Loudspeaker arrangements usually comprise a plurality of different components. A loudspeaker enclosure usually accommodates one or more loudspeakers. The loudspeaker enclosure may be mounted to a wall or, e.g., to a panel in a passenger compartment of a vehicle. The loudspeaker enclosure often is screwed to a wall or a panel, for example. Due to the movement of the loudspeaker membranes, magnets, or any other movable elements within a loudspeaker, other

elements such as a loudspeaker enclosure may also be excited and vibrate. Further, different parts and elements of the loudspeaker arrangement may be excited and bump or grate against each other. Vibrations of the loudspeaker arrangement may further be transferred to other parts and elements that are directly or indirectly connected to the loudspeaker arrangement such as, e.g., wall panels or other elements that are arranged close to the loudspeaker arrangement in a vehicle. This may result in unwanted noise which may worsen the sound experience for a user. Further, loudspeaker arrangements that are arranged in a vehicle or in a wall of a listening environment are usually required to be comparably small.

SUMMARY

(3) A loudspeaker arrangement includes a first loudspeaker including a first sound radiating surface, a second loudspeaker including a second sound radiating surface, and an enclosure. The first loudspeaker is mechanically coupled to the second loudspeaker, the enclosure encloses the second side of the first sound radiating surface and the second side of the second sound radiating surface, the enclosure comprises a port coupling an internal volume inside the enclosure to outside air, and either a first side of the first sound radiating surface faces a first side of the second sound radiating surface, and the first side of the first sound radiating surface and the first side of the second sound radiating surface both face a first cavity, the first cavity comprising a first opening facing towards an internal air volume, or a second side of the first sound radiating surface faces a second side of the second sound radiating surface, the first side of the first sound radiating surface faces a second cavity, the second cavity comprising a second opening facing towards the internal air volume, and the first side of the second sound radiating surface faces a third cavity separate and distinct from the second cavity, the third cavity comprising a third opening facing towards the internal air volume.

(4) Other systems, methods, features and advantages will be or will become apparent to one with skill in the art upon examination of the following detailed description and figures. It is intended that all such additional systems, methods, features, and advantages be included within this description, be within the scope of the invention and be protected by the following claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The arrangement may be better understood with reference to the following description and drawings. The components in the figures are not necessarily to scale, emphasis instead being placed upon illustrating the principles of the invention. Moreover, in the figures, like referenced numerals designate corresponding parts throughout the different views.

(2) FIG. **1** schematically illustrates a cross-sectional view of a loudspeaker arrangement;

(3) FIG. **2** schematically illustrates a dimensional view of another loudspeaker arrangement;

(4) FIG. **3** schematically illustrates a dimensional view of loudspeakers of a loudspeaker arrangement in an unmounted state;

(5) FIG. **4** schematically illustrates a front view of loudspeakers of a loudspeaker arrangement;

(6) FIG. **5** schematically illustrates a side view of loudspeakers of a loudspeaker arrangement;

(7) FIG. **6** schematically illustrates an exploded view of an exemplary loudspeaker arrangement;

(8) FIG. **7** schematically illustrates a cross-sectional view of an exemplary loudspeaker arrangement;

(9) FIG. **8** schematically illustrates a cross-sectional view of another exemplary loudspeaker arrangement;

(10) FIG. **9** schematically illustrates a loudspeaker arrangement comprising an uneven number of loudspeakers according to one example; and

(11) FIG. **10** schematically illustrates a loudspeaker arrangement comprising an uneven number of

loudspeakers according to another example.

DETAILED DESCRIPTION

(12) Referring to FIG. **1**, a loudspeaker arrangement **100** is schematically illustrated. In particular, FIG. **1** schematically illustrates a cross-sectional view of a loudspeaker arrangement **100**. The loudspeaker arrangement **100** may, e.g., be mounted into a wall or in a vehicle. The loudspeaker arrangement **100** comprises a first loudspeaker **110** and a second loudspeaker **120**. The first loudspeaker **110** comprises a first sound radiating surface (e.g., a first membrane), and the second loudspeaker **120** comprises a second sound radiating surface (e.g., a second membrane). The first loudspeaker **110** and the second loudspeaker **120** are arranged opposite each other in a first direction y. That is, a first side of the first sound radiating surface is arranged opposite to and faces a first side of the second sound radiating surface. The first sound radiating surface in this example is arranged essentially parallel to the second sound radiating surface. A distance d**1** between the first loudspeaker **110** and the second loudspeaker **120** may be between 1 cm and 20 cm, for example. The distance d**1** between the first loudspeaker **110** and the second loudspeaker **120** may depend on the size of the loudspeakers **110**, **120**, for example.

(13) A first cavity **130** is formed between a front side of the first loudspeaker **110** and a front side of the second loudspeaker **120**. The first loudspeaker **110** and the second loudspeaker **120** may radiate sound into the first cavity **130**. The first cavity **130** may comprise an opening through which sound generated by the first loudspeaker **110** and sound generated by the second loudspeaker **120** may exit the first cavity **130** towards the environment. The opening may be completely open or may be covered with a vented covering such as a fabric or mesh that may provide a decorative exposure to a wall containing the enclosure. A wall surface opposite the opening is closed along the entire length of the opening so that sound radiates from the opening. The opening, however, is not visible in the cross-sectional view of FIG. **1** as it is arranged in another image plane.

(14) The loudspeaker arrangement **100** may further comprise an enclosure comprising a first enclosure part **114** and a second enclosure part **124**. The first enclosure part **114** forms a first cavity **112** surrounding a back side of the first loudspeaker **110**. The second enclosure part **124** forms a second cavity **122** surrounding the back side of the second loudspeaker **120**. According to another example, the first enclosure part **114** and the second enclosure part **124** together may form a single cavity surrounding the back side of both the first loudspeaker **110** and the second loudspeaker **120**.

(15) The loudspeaker arrangement **100** illustrated in FIG. **1** comprises one pair of loudspeakers, the pair of loudspeakers including the first loudspeaker **110** and the second loudspeaker **120**. This, however, is only an example. It is also possible that a loudspeaker arrangement comprises more than two loudspeakers. This is exemplarily illustrated in FIG. **2** which schematically illustrates a dimensional view of another loudspeaker arrangement **200**. The loudspeaker arrangement **200** illustrated in FIG. **2** comprises three pairs of loudspeakers, each pair of loudspeakers formed by a first loudspeaker **210** and a second loudspeaker (not visible in the dimensional view of FIG. **2**). Any other number of loudspeakers, however, is also possible. While in the Figures only pairs of loudspeakers are illustrated, it is generally also possible that the loudspeaker arrangement **100**, **200** comprises an uneven number of loudspeakers. This will be exemplarily illustrated by means of FIGS. **9** and **10** below.

(16) Each of the at least two loudspeakers may be arranged similarly to what has been described with respect to the first and second loudspeaker **110**, **120** of FIG. **1** above. The first cavity **230** may be formed continuously between all (pairs of) loudspeakers, for example. The first enclosure part **214** may enclose all first loudspeakers **210** and form a continuous closed cavity at the back side of the first loudspeakers **210**. However, it is also possible that a separate cavity is formed at the back side of each of the first loudspeakers **210**. The same applies for the second loudspeakers and the second enclosure part **224**, which may form a single continuous cavity at the back side of the second loudspeakers, or separate cavities for each of the second loudspeakers. According to another example, it is also possible that the first loudspeakers **210** and the second loudspeakers are all

arranged in a single continuous cavity formed by the enclosure at the back sides of the loudspeakers. The opening **232** of the first cavity **230** may face towards a second direction z which is perpendicular to the first direction y. If the loudspeaker arrangement **200** comprises more than two loudspeakers, two or more loudspeakers may be arranged successively in a third direction x, which is perpendicular to both the first direction y and the second direction z.

(17) Now referring to the exploded view of FIG. **3**, a first loudspeaker **310** and a second loudspeaker **320** are schematically illustrated in an unmounted state. A loudspeaker enclosure is not specifically illustrated in FIG. **3**. The first loudspeaker **310** comprises a first loudspeaker basket **316**, and the second loudspeaker **320** comprises a second loudspeaker basket **326**. According to one example, the first loudspeaker **310** and the second loudspeaker **320** may be arranged opposite each other in the first direction y and may be mechanically coupled to each other. According to one example, the first loudspeaker basket **316** is mechanically coupled to the second loudspeaker basket **326**. The first loudspeaker **310** or first loudspeaker basket **316** may be coupled to the second loudspeaker **320** or second loudspeaker basket **326** by means of first and second connection elements **352**, **354**. For example, the first loudspeaker **310** may comprise a plurality of first connection elements **352** and a plurality of second connection elements **354**. The second loudspeaker **320** may also comprise a plurality of first connection elements **352** and a plurality of second connection elements **354**, wherein each first connection element **352** of the second loudspeaker **320** forms a counterpart for a second connection element **354** of the first loudspeaker **310**, and each second connection element **354** of the second loudspeaker **320** forms a counterpart for a first connection element **352** of the first loudspeaker **310**. Using connection elements to couple the first loudspeaker **310** to the second loudspeaker **320**, however, is only an example. The loudspeakers **310**, **320** can be coupled in any other suitable way. According to another example, the first loudspeaker **310** and the second loudspeaker **320** are glued together.

(18) The first loudspeaker **310** may comprise a first projection **318**. The first projection **318** may extend from the first loudspeaker basket **316** in the first direction y towards the second loudspeaker **320**. The first projection **318**, in a plane defined by the second direction z and the third direction x, may at least partly surround the first sound radiating surface. As is schematically illustrated in FIG. **3**, the first projection **318** may be omitted towards one side, in order to form the opening **332**. The second loudspeaker **320** may comprise a second projection **328**. The second projection **328** may extend from the second loudspeaker basket **326** in the first direction y towards the first loudspeaker **310**. The second projection **328**, in a plane defined by the second direction z and the third direction x, may at least partly surround the second sound radiating surface. As is schematically illustrated in FIG. **3**, the second projection **328** may be omitted towards one side, in order to form the opening **332**. The first connection elements **352** and the second connection elements **354** may be arranged along the first projection **318** and the second projection **328**, respectively. When the first loudspeaker **310** and the second loudspeaker **320** are coupled to each other, the first projection **318** and the second projection **328** may be coupled to each other, thereby connecting the first loudspeaker basket **316** to the second loudspeaker basket **326**. The first cavity **330** may be defined by the first and the second projection **318**, **328**. That is, the first and the second projection **318**, **328** partially surround the first cavity **330** in a plane defined by the second direction z and the third direction x.

(19) When a connection is formed between the first loudspeaker **310** and the second loudspeaker **320**, the first cavity **330** is formed between the first loudspeaker **310** and the second loudspeaker **320**, with an opening **332** formed towards the second direction z.

(20) The first loudspeaker **310** and the second loudspeaker **320** during use (e.g., when the first sound radiating surface and the second sound radiating surface are excited in order to produce sound) both generate vibrations. By mechanically connecting the first loudspeaker **310** to the second loudspeaker **320**, the vibrations of the two loudspeakers **310**, **320** can cancel each other out. That is because both loudspeakers **310**, **320** generally receive the same sound signal and produce

the same sound at the same time. That is, both loudspeakers vibrate simultaneously. By mounting the loudspeakers to face each other, the vibrations are inverse to each other and, therefore, counteract each other. In this way, the resulting vibrations of the loudspeaker arrangement **300** comprising the first loudspeaker **310** and the second loudspeaker **320**, are zero or at least close to zero. Therefore, almost zero vibration is transferred to any surround parts such as a loudspeaker enclosure, for example. The same applies for an arrangement comprising more than two loudspeakers.
(21) A first loudspeaker **410** and a second loudspeaker **420** in a mounted state are exemplarily illustrated in the front view of FIG. **4** and in the side view of FIG. **5**. As can be seen in the side view of FIG. **5** for example, when the first loudspeaker **510** is connected to the second loudspeaker **520**, a third projection **540** and a fourth projection **544** are formed by the first loudspeaker **510** and the second loudspeaker **520**. The third projection **540** extends from the loudspeaker arrangement **500** in the second direction z, and the fourth projection **544** extends from the loudspeaker arrangement **500** in the second direction z, opposite to the first projection **540**. According to one example, the third projection **540** may be formed by the first loudspeaker basket **516** or the first projection **518** and the second loudspeaker basket **526** or the second projection **528**, and the fourth projection **544** may also be formed by the first loudspeaker basket **516** or the first projection **518** and by the second loudspeaker basket **526** or the second projection **528**. For example, an upper half of the third projection **540** and an upper half of the fourth projection **544** may be formed by the first loudspeaker **510** (first loudspeaker basket **516** or first projection **518**), and a lower half of the third projection **540** and a lower half of the fourth projection **544** may be formed by the second loudspeaker **520** (second loudspeaker basket **526** or second projection **528**).
(22) Each of the third projection **440**, **540** and the fourth projection **444**, **544** may comprise a protruding edge or ledge, for example. The third projection **440**, **540** may surround the opening **432** of the first cavity **430** in a plane defined by the second and third direction z, x, for example. As is schematically illustrated in FIG. **4**, the opening **432** may have an elongated form (cross-section), e.g., rectangular with rounded corners. Any other form (cross-section) of the opening **432**, however, is also possible such as, square, rectangular, rounded, or oval, for example. Again referring to FIG. **5**, the fourth projection **544** may be arranged opposite to the third projection **540** in the second direction z. That is, the fourth projection **544** may be arranged at a rear wall of the first cavity **530** and the third projection **540** may be arranged at the front of the first cavity **530**, for example. This, however, is only an example. According to another example (not illustrated), the third projection **540** may be arranged at a first side wall of the first cavity **530** and the fourth projection **544** may be arranged at a second side wall of the first cavity **430**, for example. In the latter case, however, the third projection **540** may not surround the opening **532** of the first cavity **530**. The third projection **540** and the fourth projection **544** can be used, for example, to mount the loudspeakers **410**, **420** in an enclosure (enclosure not illustrated in FIGS. **4** and **5**).
(23) Now referring to FIG. **6**, a loudspeaker arrangement according to one example is schematically illustrated. In the example illustrated in FIG. **6**, the first loudspeaker **610** (i.e., the first loudspeaker basket) and the second loudspeaker **620** (i.e., the second loudspeaker basket) are mechanically coupled to each other, as has been explained with respect to FIGS. **1** to **5** above. The loudspeaker enclosure **634** illustrated in FIG. **6** comprises a cavity or air volume chamber **690** and a vent duct or port **692**. The port **692** allows to vent the cavity **690** to outside air. That is, the opening **632** of the first cavity **630** formed between the first loudspeaker **610** and the second loudspeaker **620** may face the interior of a vehicle or a room. The loudspeakers **610**, **620**, therefore, radiate sound into the internal air volume (e.g., room, or car cabin), while the port **692** provides a vent towards the outside air (e.g., outside of room, or outside of vehicle). This allows for a small size of the loudspeaker arrangement. In particular, the enclosure **634** as illustrated in FIG. **6** can be much smaller than an enclosure not comprising a port. By providing an enclosure **634** comprising a port **692**, the enclosure **634** provides an unclosed air volume inside the cavity **690** which

communicates with the outside air (e.g., with the outside of a vehicle). In this way, antiphase sound pressure generated at the rear sides of the loudspeakers **610**, **620** (rear sides facing the cavity **690**) is prevented from entering the internal air volume (e.g., room, or car cabin), thus preventing sound cancellation in the internal air volume. This can prevent a reduction in sound pressure in the internal air volume. Further, a degradation of sound quality can be reduced.

(24) The arrangement comprising the first loudspeaker **610** and the second loudspeaker **620** and, optionally, further loudspeakers, that has been described with respect to FIGS. **1** to **5** above, may be arranged in the enclosure **634** such that a single cavity **690** is formed, the single cavity **690** surrounding the back side of the at least two loudspeakers. The loudspeaker arrangement may further comprise a cover **694**, configured to close the enclosure **634** towards the internal air volume. That is, the cover **694** may form a part of a panel in the interior of a vehicle or a part of a wall in a room, for example. The cover **694** comprises an opening **696**. The opening **696** may be aligned with the opening **632** of the first cavity **630**. In this way, the opening **632** of the first cavity **630** remains open and uncovered, while the enclosure **634** and, in particular, the air volume chamber **690** is closed towards the internal air volume. That is, the enclosure **634** is closed towards the internal air volume, while an opening is provided by the port **692** towards outside air.

(25) Now referring to FIG. **7**, a cross-sectional view of the arrangement described with respect to FIG. **6** is schematically illustrated. FIG. **7** illustrates the first loudspeaker **710** and the second loudspeaker **720** facing each other (first side of first sound radiating surface faces first side of second sound radiating surface), with a first cavity **730** formed between the first loudspeaker **710** and the second loudspeaker **720**. As has been described with respect to FIG. **3** above, the first loudspeaker **710** (i.e., the first speaker basket) and the second loudspeaker **720** (i.e., the second speaker basket) are mechanically coupled to each other. The first loudspeaker **710** (or first loudspeaker basket) can be connected to the second loudspeaker **720** (or second loudspeaker basket) by means of connection elements, as has been described with respect to FIG. **3** above. It is also possible to glue the first and second loudspeakers **710**, **720** together. The first loudspeaker **710** and the second loudspeaker **720** are mounted in an enclosure **734** which forms a back cavity **790** enclosing the back side of the first loudspeaker **710** and the back side of the second loudspeaker **720** (second side of first sound radiating surface and second side of second sound radiating surface face towards back cavity **790**. The enclosure **734** comprises a port **792** configured to couple the back cavity **790** to outside air. The loudspeaker arrangement **700** can be mounted to a panel **798** of a vehicle, for example, the panel **798** separating the internal air volume from outside air. The loudspeaker arrangement **700** can comprise a cover **794** which closes the enclosure **734** or back cavity **790** towards the internal air volume. The cover **794** comprises an opening, as has been described with respect to FIG. **6** above, which is aligned with the opening **732** of the first cavity **730**.

(26) The first loudspeaker **710** and the second loudspeaker **720** and, optionally, additional loudspeakers all facing towards a single cavity **730** formed between a first side of a first sound radiating surface (first loudspeaker **710**) and a first side of a second sound radiating surface (second loudspeaker **720**), however, is only an example. According to another example and as schematically illustrated in FIG. **8**, it is alternatively possible that a second side of the first sound radiating surface is arranged opposite to and faces a second side of the second sound radiating surface. Similar to the arrangements described with respect to FIGS. **1** to **7** above, in the example of FIG. **8**, the first sound radiating surface is arranged essentially parallel to the second sound radiating surface, and the first loudspeaker **810** (i.e., the first loudspeaker basket) is mechanically coupled to the second loudspeaker **820** (i.e., the second loudspeaker basket) in order to cancel vibrations generated by each of the two loudspeakers **810**, **820** out. However, in the example illustrated in FIG. **8**, each of the first loudspeaker **810** and the second loudspeaker **820** faces a cavity **830**a, **830**b distinct and separate from the respective other cavity. That is, the first loudspeaker **810** (first side of first sound radiating surface) faces a second cavity **830**a, and the

second loudspeaker **820** (first side of second sound radiating surface) faces a third cavity **830***b*, distinct and separate from the second cavity **830***a*. The second cavity **830***a* comprises a second opening **832***a* towards the internal air volume, and the third cavity **830***b* comprises a third opening **832***b* towards the internal air volume.

(27) The first loudspeaker **810** and the second loudspeaker **820** are arranged in an enclosure **834** which forms a back cavity **890** surrounding the back side of the first loudspeaker **810** (second side of first sound radiating surface) and the back side of the second loudspeaker **820** (second side of second sound radiating surface). The enclosure **834** comprises a port **892**, similar to the arrangements described with respect to FIGS. **6** and **7** above, which couples the back cavity **890** to outside air. The first loudspeaker **810** and the second loudspeaker **820** are mechanically coupled to each other, similar to what has been explained above. According to one example, a first loudspeaker basket of the first loudspeaker is coupled to a second loudspeaker basket of the second loudspeaker **829** along a connecting line, indicated with a dot-dashed line in FIG. **8**. The loudspeakers **810**, **820** can either be coupled to each other by means of first and second connection elements, as has been described with respect to FIG. **3** above. It is, however, also possible to glue the loudspeakers **810**, **820** together along the connection line, or to use any other suitable kind of connection elements. The loudspeaker arrangement **800** can further comprise a cover **894**, wherein the cover comprises two openings, wherein each of the two openings of the cover **894** is aligned with a different one of the second opening **832***a* of the second cavity **830***a*, and the third opening **832***b* of the third cavity **830***b*.

(28) The port **892** in FIG. **8** is arranged centrally between the first loudspeaker **810** and the second loudspeaker **820**. This, however, is only an example. The port **892** can be arranged at any suitable position of the enclosure **834**. Generally, the exact position of the port **892** can depend on the design of a vehicle or a wall to which the loudspeaker arrangement **800** is mounted. The loudspeaker arrangement **800** can be mounted to a panel **898** of a vehicle, for example, the panel **898** separating the internal air volume from outside air.

(29) As has been described above, a loudspeaker arrangement comprises two or more loudspeakers. A first loudspeaker and a second loudspeaker can be arranged such that their membranes or sound radiating surfaces are (mechanically) arranged parallel to each other. According to one example, the first side of the first sound radiating surface of the first loudspeaker faces a first side of the second sound radiating surface of the second loudspeaker (see, e.g., FIG. **7**). According to another example, the second side of the first sound radiating surface of the first loudspeaker faces a second side of the second sound radiating surface of the second loudspeaker (see, e.g., FIG. **8**). Additional loudspeakers can be arranged such that their sound radiating surface is aligned with either the sound radiating surface of the first loudspeaker, or with the sound radiating surface of the second loudspeaker. That is, the sound radiating surface of each of at least one additional loudspeaker is arranged in the same plane as either the sound radiating surface of the first loudspeaker or the sound radiating surface of the second loudspeaker. According to one example, the loudspeaker arrangement comprises an even number of loudspeakers, wherein two loudspeakers together form a pair of loudspeakers, respectively. The pairs of loudspeakers can be arranged adjacent to each other in one row, similar to what has been described with respect to FIG. **3** above. In an arrangement as has been described with respect to FIG. **7** above, a first side of the sound radiating surface of each of the additional loudspeakers faces the first cavity **730**. In an arrangement as has been described with respect to FIG. **8** above, a first side of a sound radiating surface of a first loudspeaker of each pair of loudspeakers faces the second cavity **830***a*, and a first side of a sound radiating surface of a second loudspeaker of each pair of loudspeakers faces the third cavity **830***b*.

(30) Now referring to FIG. **9**, a loudspeaker arrangement **900** comprising an uneven number of loudspeakers according to one example is schematically illustrated. The loudspeaker arrangement **900** in this example comprises three loudspeakers **910**, **920**, **980**. The second sides of the sound radiating surfaces of each of the loudspeakers **910**, **920**, **980** are oriented towards each other. The

first sides of the sound radiating surfaces of each of the loudspeakers **910**, **920**, **980** are positioned away from each other. Each of the three loudspeakers **910**, **920**, **980** are arranged at an equal angle **97** from each other in a radial array. As is illustrated in FIG. **9**, the three loudspeakers **910**, **920**, **980** are oriented so that the central axes C, D, E of the acoustic radiation are oriented at generally 120 degrees from each other. The acoustic radiation C, D, E from the second side of the sound radiating surfaces intersects at the circumcenter **98**. The loudspeakers **910**, **920**, **980** are arranged in an enclosure **934**. The enclosure **934** in this example is shaped as a hexagon with one of the loudspeakers mounted on alternating sides of the hexagon. However, the enclosure **934** could also be triangular, for example, with one of the loudspeakers **910**, **920**, **980** mounted on each side of the enclosure **934**. Each of the loudspeakers **910**, **920**, **980** (first side of each sound radiating surface) faces a separate cavity **930***a*, **930***b*, **930***c*. The loudspeakers **910**, **920**, **980** are mechanically coupled to each other.

(31) FIG. **10** illustrates another exemplary loudspeaker arrangement **1000** comprising an uneven number of loudspeakers **1010**, **1020**, **1080**. In this example, the first sides of the sound radiating surfaces of the loudspeakers **1010**, **1020**, **1080** are oriented toward each other. The second sides of the sound radiating surfaces of the loudspeakers **1010**, **1020**, **1080** are oriented away from each other. Each of the three loudspeakers **1010**, **1020**, **1080** are arranged at an equidistant angle **107** from each other in a radial array. As is illustrated in FIG. **10**, the three transducers **1010**, **1020**, **1080** are oriented so that the central axes J, K, L of the acoustic radiation are angularly equidistant from each other. In this example with three loudspeakers **1010**, **1020**, **1080**, the central axes J, K, L are oriented at generally 120 degrees from each other. The acoustic radiation J, K, L from the first sides of the sound radiating surfaces intersect at the circumcenter **108** of a cavity **1030**. The loudspeakers **1010**, **1020**, **1080** can be arranged in an enclosure (not specifically illustrated in FIG. **10**) which encloses the second sides of the sound radiating surfaces of the loudspeakers **101**, **1020**, **1080**. In the example illustrated in FIG. **10**, the cavity **1030** has a triangular form. This, however, is only an example. Other forms are generally possible. The loudspeakers **1010**, **1020**, **1080** are mechanically coupled to each other.

(32) The first, second and third direction x, y, z are used throughout the description for illustrative purposes only. The directions are not meant to be defined with regard to a ground surface. Rather, the directions are merely used to illustrate the orientation of the different elements and their arrangement with regard to each other. When mounted into a wall or inside a vehicle, the described loudspeaker arrangement may be rotated into any suitable position resulting in any suitable orientation.

(33) While various embodiments of the invention have been described, it will be apparent to those of ordinary skill in the art that many more embodiments and implementations are possible within the scope of the invention. In particular, the skilled person will recognize the interchangeability of various features from different embodiments. Although these techniques and systems have been disclosed in the context of certain embodiments and examples, it will be understood that these techniques and systems may be extended beyond the specifically disclosed embodiments to other embodiments and/or uses and obvious modifications thereof. Accordingly, the invention is not to be restricted except in light of the attached claims and their equivalents.

(34) The description of embodiments has been presented for purposes of illustration and description. Suitable modifications and variations to the embodiments may be performed in light of the above description or may be acquired from practicing the methods. The described arrangements are exemplary in nature, and may include additional elements and/or omit elements. As used in this application, an element recited in the singular and proceeded with the word “a” or “an” should be understood as not excluding plural of said elements, unless such exclusion is stated. Furthermore, references to “one embodiment” or “one example” of the present disclosure are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features. The terms “first,” “second,” and “third,” etc. are used merely as labels, and are not

intended to impose numerical requirements or a particular positional order on their objects. The described systems are exemplary in nature, and may include additional elements and/or omit elements. The subject matter of the present disclosure includes all novel and non-obvious combinations and sub-combinations of the various systems and configurations, and other features, functions, and/or properties disclosed. The following claims particularly point out subject matter from the above disclosure that is regarded as novel and non-obvious.

Claims

1. A loudspeaker arrangement comprising: a first loudspeaker having a first sound radiating surface; a second loudspeaker having a second sound radiating surface; an enclosure (**734**, **834**); the first loudspeaker is mechanically coupled to the second loudspeaker; the enclosure encloses the second side of the first sound radiating surface and the second side of the second sound radiating surface; the enclosure has a port coupling a back cavity inside the enclosure to outside air; and either a first side of the first sound radiating surface faces a first side of the second sound radiating surface, and the first side of the first sound radiating surface and the first side of the second sound radiating surface both face a first cavity, the first cavity has a first opening facing towards an internal air volume; or a second side of the first sound radiating surface faces a second side of the second sound radiating surface, the first side of the first sound radiating surface faces a second cavity, the second cavity has a second opening (**832***a*) facing towards the internal air volume, and the first side of the second sound radiating surface faces a third cavity separate and distinct from the second cavity, the third cavity has a third opening facing towards the internal air volume.
2. The loudspeaker arrangement of claim 1, wherein the sound radiating surface of the first loudspeaker is arranged parallel to the sound radiating surface of the second loudspeaker.
3. The loudspeaker arrangement of claim 1, further comprising at least one additional loudspeaker having a third sound radiating surface; either a first side of the third sound radiating surface faces the first sides of the first sound radiating surface and the second sound radiating surface, and faces the first cavity (; or a second side of the third sound radiating surface faces the second sides of the first sound radiating surface and the second sound radiating surface, and faces a fifth cavity separate and distinct from the first cavity and the second cavity, the fifth cavity has a fifth opening facing towards the internal air volume.
4. The loudspeaker arrangement of claim 1, wherein; the loudspeaker arrangement has an even number of loudspeakers; two loudspeakers together form a pair of loudspeakers; the pairs of loudspeakers are arranged in one row; and either a first side of the sound radiating surface of each of the loudspeakers in the pair of loudspeakers faces the first cavity; or a first side of a sound radiating surface of a first loudspeaker of each pair of loudspeakers faces the second cavity, and a first side of a sound radiating surface of a second loudspeaker of each pair of loudspeakers faces the third cavity.
5. The loudspeaker arrangement of claim 1, further comprising a cover configured to close the enclosure towards the internal air volume.
6. The loudspeaker arrangement of claim 5, wherein; either the cover has an opening aligned with the first opening of the first cavity; or the cover has two openings each of the two openings of the cover is aligned with a different one of the second opening of the second cavity and the third opening of the third cavity.
7. The loudspeaker arrangement of claim 5, wherein; either the cover forms a part of a panel in the interior of a vehicle; or the cover forms a part of a wall in a room.
8. The loudspeaker arrangement of claim 1, wherein; either the first opening is covered with a fabric or mesh; or the second opening and the third opening are covered with a fabric or mesh.
9. The loudspeaker arrangement of claim 1, wherein the first loudspeaker has a first speaker basket, the second loudspeaker has a second speaker basket, and the first speaker basket is mechanically

coupled to the second speaker basket.

10. The loudspeaker arrangement of claim 9, further comprising a plurality of first connection elements and a plurality of second connection elements configured to connect the first speaker basket to the second speaker basket.

11. The loudspeaker arrangement of claim 9, wherein the first speaker basket and the second speaker basket are glued together.

12. The loudspeaker arrangement of claim 9, wherein; a first side of the first sound radiating surface faces a first side of the second sound radiating surface, and the first side of the first sound radiating surface and the first side of the second sound radiating surface both face a first cavity, the first cavity has a first opening facing towards an internal air volume; the first speaker basket has a first projection extending from the first speaker basket in a first direction (y) towards the second loudspeaker; the second speaker basket has a second projection extending from the second speaker basket in the first direction (y) towards the first loudspeaker; and the first projection is mechanically coupled to the second projection defining the first cavity between a front side of the first loudspeaker and a front side of the second loudspeaker.

13. The loudspeaker arrangement of claim 12, wherein; the first projection and the second projection are omitted in a second direction (z) that is perpendicular to the first direction (y), in order to form the first opening.

14. The loudspeaker arrangement of claim 1, wherein; either the first opening has a rectangular cross-section with rounded corners, a square cross-section, a rectangular cross-section, a rounded cross-section, or an oval cross-section; or each of the second opening and the third opening has a rectangular cross-section with rounded corners, a square cross-section, a rectangular cross-section, a rounded cross-section, or an oval cross-section.

15. The loudspeaker arrangement of claim 1, wherein the loudspeaker arrangement is mounted to a panel of a vehicle, the panel separating the internal air volume from outside air.